

Postability Scorer

A personalized model that learns your Instagram aesthetic and recommends photos from your camera roll to post.

The Problem & Goal

The Problem

Thousands of photos sit on your camera roll.

Only a handful ever get posted to Instagram.

Scrolling through everything to pick the right one is slow, annoying, and inconsistent.

The Goal

Build a model that learns your personal aesthetic from your past Instagram posts...

...and ranks camera roll photos by how likely you are to post them.

The key insight: Why I rebuilt the model?

v1

Binary classifier

Instagram photos = postable (1)

Camera roll = not postable (0)

The flaw: camera roll photos aren't 'bad' - they're just unposted.

Result: only 2/10 recommendations felt right.



v2 (This Version)

Similarity-based ranking

Builds a taste profile from your IG posts (global centroid + aesthetic clusters).

Scores camera roll by cosine similarity to that profile, combined with quality features.

Result: 6-7/10 recommendations felt right.

How It Works

1 CLIP Embeddings

Both IG posts and camera roll photos are embedded using CLIP (ViT-B/32) - a vision model that maps images to a 512-dim semantic space.

2 Taste Profile

Your Instagram posts form a global taste centroid. KMeans (k=4) also finds aesthetic sub-clusters in your posts.

3 Quality Features

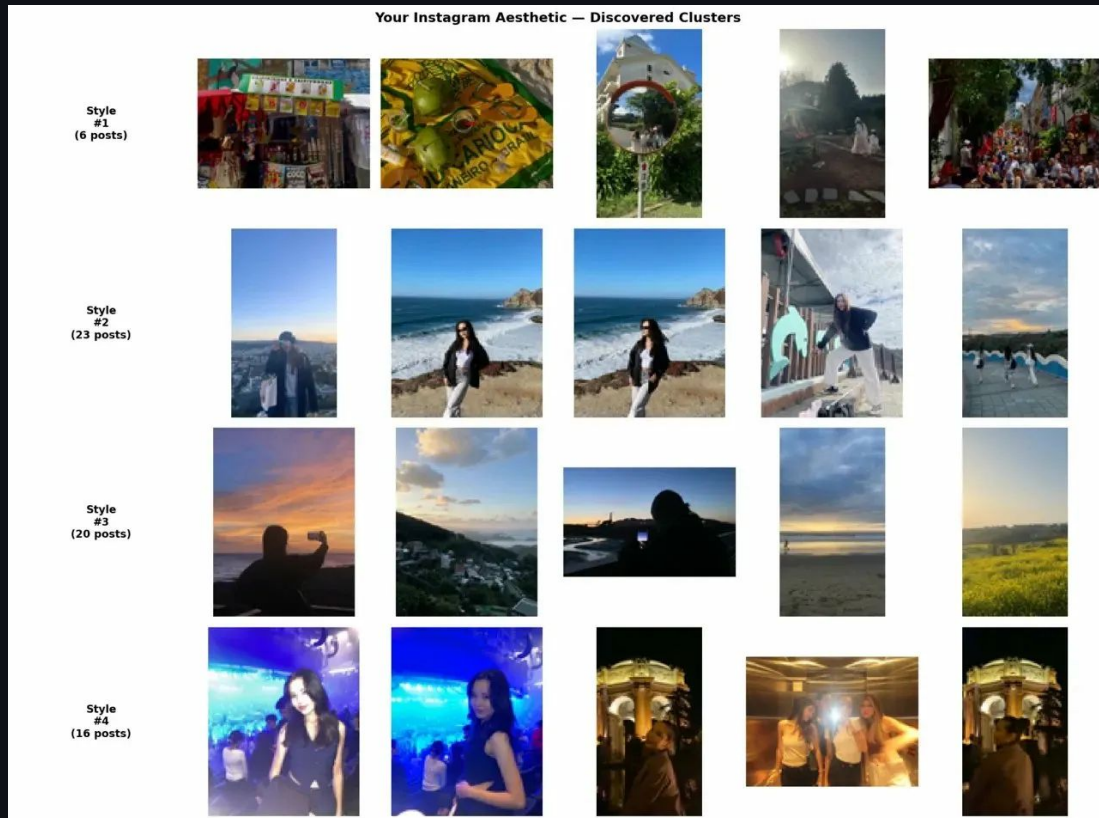
6 engineered features per photo: sharpness, brightness, contrast, saturation, face detection, center composition.

4 Final Score

Score = $0.65 \times \text{style similarity} + 0.35 \times \text{quality score}$. Quality is normalized to YOUR posting standards, not a universal benchmark.

Discovering my aesthetic - 4 clusters

KMeans found 4 visually distinct posting styles in my Instagram history:



Cluster

Your
Style #3
Posts
(20 total)



18006534485397384



18012685643615146



18017456243241576



18100469943411003

score: 0.71



score: 0.71



score: 0.69



score: 0.69



* Model
Recommends
for Style #3
(67 candidates)

Recommendation

Cluster

Your
Style #2
Posts
(23 total)



score: 0.78



score: 0.77



score: 0.73



18453238462003292



* Model
Recommends
for Style #2
(74 candidates)



score: 0.69

Recommendation

Top 10 recommendations - v2 model

Top 10 Recommended Photos from Your Camera Roll (v2 Model)

#1 score: 0.80
style: 0.98 quality: 0.48



#2 score: 0.80
style: 0.97 quality: 0.48



#3 score: 0.78
style: 0.91 quality: 0.53



#4 score: 0.77
style: 0.93 quality: 0.48



#5 score: 0.77
style: 0.91 quality: 0.49



#6 score: 0.74
style: 0.87 quality: 0.50



#7 score: 0.73
style: 0.80 quality: 0.60



#8 score: 0.73
style: 0.85 quality: 0.50



#9 score: 0.72
style: 0.83 quality: 0.52



#10 score: 0.71
style: 0.70 quality: 0.72



Evaluation

60%

Precision@10

6 out of 10 I'd actually post

60%

Precision@20

12 out of 20 I'd actually post

~20%

v1 baseline

2 out of 10 felt right

How I evaluated:

Since there are no ground-truth labels for the camera roll, I built a self-evaluation loop: walked through the top 20 recommendations one by one and recorded whether I would actually post each one. This gives a Precision@K metric. I also tried training a re-ranking layer (logistic regression on my labeled data) but with only 20 training examples it overfit badly: v2's 60% beat v3's 25%, so I stayed with v2.

Limitations & What's Next

Limitations

Visual only

No caption, mood, or timing context.

Taste shift

Older posts may not reflect current aesthetic (but old photos still trained the model.)

If I Had More Time

More labels

Label 100+ photos to properly train a re-ranker like LTR or XGBoost.

Recency weighting

Discount older posts so the model tracks taste evolution.

App wrapper

Package it as a mobile tool that can read a real camera roll and return a ranked shortlist.